

This is sample data provided for CALIBER purposes only		ONE POINT LESSON (OPL)		OPL No: OPL-KC-4501-02 Sheet 1 of 1
OPL Title	Compressor Suction/Discharge Valve Inspection	Discipline	Mechanical	
Equipment	KC-4501 - RECYCLE GAS COMPRESSOR	Area / Unit	4500 - RECYCLE GAS SYSTEM	
Related Interlock	SEQ-4501 (SIL 2)	P&ID Ref	TJC-LLD-PID-4501	
Classification:	☐ Basic Knowledge	☐ Improvement	☒ Trouble Case	Aspect: P-Q-C-D-S-M-E

1. PURPOSE / OBJECTIVE

This One Point Lesson explains **compressor suction/discharge valve inspection** for KC-4501 (RECYCLE GAS COMPRESSOR) in the RECYCLE GAS SYSTEM. Broken plate valves are the classic cause of high discharge temperature (TSHH-4503) and lost capacity on KC-4501. The lesson gives operators and technicians the key knowledge and the correct step-by-step method so the task is done safely, correctly and consistently every time.

2. SAFETY PRECAUTIONS

Hazard note: this equipment is **HIGH CRITICAL** handling **Ethylene/Hexane/H2** at up to 16 barg / 150 degC. Observe the following before and during the task:

- Apply LOTO (Lock-Out Tag-Out) and prove zero energy before starting work.
- Isolate, depressurise, drain and purge the equipment; obtain a valid work permit.
- Wear the correct PPE: coverall, safety shoes, gloves, goggles and hearing protection.
- Beware of stored mechanical energy (springs, rotating inertia) and pinch points.
- Use certified lifting gear and correct manual-handling technique for heavy parts.

3. TOOLS & MATERIALS REQUIRED

- Standard mechanical hand tools & calibrated torque wrench
- LOTO padlocks and permit
- Laser/dial alignment kit (as applicable)
- Feeler gauges, dial indicator, seal/bearing pullers
- Correct spare parts and gaskets per the BOM
- Lifting gear and slings (certified)

4. DETAILED PROCEDURE / STEPS

Step	Action	Check / Acceptance
1	Isolate, depressurise and purge the cylinder with nitrogen	Isolation & LOTO verified
2	Remove the suction and discharge valve caps	Parts match BOM / spec
3	Inspect valve plates, springs and seats; replace as a complete set	Torque / clearance to spec
4	Check the valve gasket and seat for damage	No leak / smooth operation
5	Torque the valve caps to spec and leak test before start	Record in equipment history

5. COMMON PROBLEMS & TROUBLESHOOTING

Symptom	Likely Cause	Action
High discharge temperature trip TSHH-4503, capacity drop	1st stage suction plate valve fractured	Replaced complete suction valve set, checked seat, leak tested
KC-4501 tripped on PSLL-4504 low lube oil pressure	Lube oil pump gears worn, relief valve set low	Overhauled lube oil pump, reset relief valve, verified 2.5 barg
High blow-by, low 2nd stage efficiency	PTFE piston rings worn beyond limit	Renewed 2nd stage PTFE piston and rider rings, honed cylinder

6. KEY LEARNING POINTS

- Broken plate valves are the classic cause of high discharge temperature (TSHH-4503) and lost capacity on KC-4501.
- Respect the interlock SEQ-4501: e.g. PSLL-4504 trips at < 1.5 barg - never defeat it without a permit.
- Always cross-check the Datasheet limits and the P&ID TJC-LLD-PID-4501 before working on KC-4501.
- Record the work in the equipment history so the Knowledge Hub stays the single source of truth.

Prepared by	Reviewed by (Supervisor)	Approved by (Manager)	Date of Sharing
Panel Operator / Technician	Wahyu Setiadi (EMP-1113)	Zulkarnain Hasan (EMP-0901)	Tuesday, 28 April 2026